

Abdullah Mahmoud Fawzy

Frontend / Fullstack Engineer

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Professional Summary

Software Engineer with a Bachelor’s degree in Mathematics from Helwan University and a strong analytical foundation in problem solving, logic, and algorithmic thinking. Specializing in modern frontend and fullstack engineering with Next.js, TypeScript, React, and Node.js. Experienced in designing maintainable web architectures using SOLID principles and Strategy/Factory patterns, building real-time systems, implementing secure authentication and request protection, and optimizing server/client data flows.

Education

Bachelor of Science in Mathematics, *Helwan University* 10/2024
Cairo, Egypt

Technical Skills

Languages

TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

UI & Styling

Tailwind CSS, CSS Modules, Responsive Design, Accessible UI (ARIA), Recharts

Databases & Cloud

PostgreSQL, Supabase, MongoDB, Mongoose, Vercel

Frontend

Next.js (App Router, Server & Client Components, Server Actions), React, React Hooks, Redux Toolkit, TanStack Query

Backend & APIs

Node.js, Express.js, RESTful APIs, WebSockets (Socket.IO), JWT Authentication, Zod Schema Validation, Rate Limiting

Architecture & Engineering

SOLID Principles, Strategy Pattern, Factory Pattern, Optimistic UI Updates, Server-Side Prefetching & Client-Side Hydration, Git & GitHub

Featured Engineering Projects

Devfolio AI, Engineering Showcase & Architectural Assistant

Next.js 15, TypeScript, Tailwind CSS, Strategy Pattern, Rate Limiting

- Architected an engineering-focused portfolio platform with dynamic case studies and an interactive AI technical assistant.
- Decoupled LLM provider integration using Strategy and Factory patterns, enabling provider-agnostic integration and simplifying the addition or replacement of providers.
- Applied Single Responsibility and Open/Closed principles to separate provider-specific logic from the application layer.
- Implemented an in-memory sliding-window rate limiter for serverless API routes to mitigate abusive traffic, excessive requests, and downstream API quota exhaustion.
- Enforced end-to-end type safety with strict TypeScript configuration and schema-based request validation.

Inventory Management Dashboard

Next.js 16, TypeScript, TanStack Query, Tailwind CSS, Recharts

- Engineered an inventory and analytics dashboard using Next.js App Router and TanStack Query.
- Implemented server-side data prefetching with client-side cache hydration, improving initial data availability and avoiding redundant client-side requests.
- Built multi-parameter filtering, sorting, debounced search, and client-side pagination across 1,000+ inventory records.
- Developed interactive analytics modules for revenue trends, low-stock alerts, and category distribution using Recharts.

Real-Time Team Chat & Collaboration Application

React, Redux Toolkit, Node.js, Express, MongoDB, Socket.IO, JWT

- Built a fullstack real-time messaging application with persistent conversations and active user presence.

- Engineered bidirectional communication using Socket.IO and room-based broadcasting, with optimistic client-side updates for responsive messaging.
- Implemented stateless JWT authentication using HTTP-only secure cookies, bcrypt password hashing, and protected Express middleware.
- Managed complex client-side conversation state with Redux Toolkit, including active conversations and unread message notifications.

Hotel Management & Operations Dashboard

React, Supabase, PostgreSQL, React Query, React Hook Form, Recharts

- Developed an internal operations dashboard for managing bookings, cabin inventory, and guest check-in/check-out workflows.
- Integrated Supabase PostgreSQL with Row Level Security (RLS) policies for controlled data access.
- Implemented structured form validation with React Hook Form to prevent invalid reservation submissions and provide clear client-side error feedback.
- Built administrative analytics covering occupancy, revenue, and stay-duration metrics.

Languages

Arabic
Native

English
Professional Working Proficiency